Can Oil Prices Improve Airline Stock Forecasts?

DS 4002 Case Study by Anish Jagota

Airline companies are constantly exposed to changes in oil prices. When oil prices rise, airlines often face higher fuel costs, which can reduce profitability. When oil prices fall, airlines may benefit from lower operating expenses. Because of this, investors often assume that oil prices should help explain or even predict airline stock performance.

But in practice, the relationship is not so simple. Airline stocks are also affected by broader market conditions, investor expectations, economic growth, and demand for travel. This raises an important question: **does oil actually improve our ability to forecast airline returns, or does the broader stock market already explain most of what matters?**

You are working as a junior data analyst for a financial research team that wants to test whether oil prices add useful forecasting information for airline stocks. Your task is to investigate this question using publicly available Yahoo Finance data, focusing on crude oil, the airline ETF **JETS**, and the market benchmark **SPY**.

Your job is to build a small forecasting study that compares different approaches to prediction. You should explore the data, test the relationship between oil and airline returns, and evaluate whether adding oil-related information improves predictive performance relative to simpler benchmark models.

By the end of the case study, you will produce a short but well-organized analysis that answers a practical investment question:

Can oil prices meaningfully improve forecasts of airline stock performance?

This project is meant to simulate the kind of work a real analyst might do when deciding whether an additional variable is actually useful in prediction. Your goal is not just to run models, but to decide whether the extra complexity is worth it.